

STM32F0 series

Entry-level Cortex™-M0 MCU



STM32  Releasing your **creativity**



STMicroelectronics

Introducing **STM32F0** with **STM32® DNA** Inside Get **32-bit** Performance at **8-bit Budget Price**

System Power supply 1.8 V regulator POR/PDR/PVD Xtal oscillators 32 kHz + 4 - 32 MHz Internal RC oscillators 40 kHz + 8 MHz PLL Clock control Calendar RTC SysTick timer 2x window watchdogs (independent and system) 27/39/55 I/Os Touch-sensing controller Cyclic redundancy check (CRC)	ARM Cortex-M0 CPU 48 MHz Nested vector interrupt controller (NVIC) SW debug AHB-Lite bus matrix APB bus 5-channel DMA	Up to 64-Kbyte Flash memory Up to 8-Kbyte SRAM HW parity checking 20-byte backup data Connectivity HDMI CEC 2x SPI (1x with PS mode) 2x I ² C (1x with fast-mode plus) 2x USART with modem control (1x with LIN, smartcard, IrDA)
Control 1x 16-bit motor control PWM Synchronized AC timer 1x 32-bit timer 5x 16-bit PWM timers 1x 16-bit basic timer		Analogs 1-channel 12-bit DAC 1x 12-bit ADC 16 channels / 1 MSPS 2x analog comparator

Key features

Core and operating conditions
 ARM® Cortex™-M0
 0.9 DMIPS/MHz up to 48 MHz
 1.8/2.0 to 3.6 V supply range

High-performance connectivity
 6 Mbit/s USART 18 Mbit/s SPI
 with 4- to 16-bit data frame
 1 Mbit/s I²C fast-mode plus HDMI CEC

Enhanced control
 16-bit 3-phase PWM motor control timer
 5x 16-bit PWM timers 16-bit basic timer
 32-bit PWM timer 12 MHz I/O toggling

Targeted applications

- Home automation
- Home appliances
- Motor control
- Sensors
- A/V receivers, TVs
- Remote controls
- Gaming

STM32F0-Discovery Kit

Everything you need to discover
 STM32F0 32-Bit ARM Cortex-M0
 based MCUs having embedded
 real time ST-Link/V2
 programmer/debugger



Please post your queries to: mcu-enquiry@st.com

www.st.com/mcu